Netradyne Campus Innovation Challenge: Fleet Route Risk Intelligence

PathPrahari

पथ-प्रहरी
(guardian of the route)

Route and departure-time risk scoring for Indian truck fleets, before the truck leaves the depot.

Team Hackzz

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Before every dispatch, PathPrahari answers:



How risky is this trip tonight?



Which stretches make it risky and why?



Is there a safer route that costs little extra time?



Would leaving at a different hour be safer?

Trucks are dispatched without anyone checking the risk

1,72,890

people died on Indian roads in 2023

4,80,583

road accidents in 2023, about 55 every hour

~36%

of road deaths are on National Highways, which are ~2% of road length

5,036 of 13,795

NH blackspots have long-term repairs done

A typical dispatch decision today

Route chosen by: shortest time or distance

Harsh-braking data: sits in the telematics dashboard, unused

Night, fog, unlit roads, tired drivers: not part of the decision

Blackspot repairs: years away for most of the list

Sources: MoRTH, Road Accidents in India 2023 (via PTI and OpenCity); Lok Sabha reply on NH/SH share of fatalities; Parliamentary Standing Committee on Transport report on blackspots.

Why road shape alone can't tell you where crashes happen

67%

of India's road accidents in 2023 were on straight roads.

Curves, potholes and steep grades together: 14.5%.

So a map of curves and blackspots misses most of the danger.

What actually changes the risk on a stretch

How vehicles behave there

Repeated harsh braking and near-misses show trouble before crashes add up.

The hour

Night driving and the 2–5 AM fatigue window raise risk on the very same road.

Light and weather

Unlit stretches, winter fog in the north and monsoon rain change risk hour by hour.

Source: MoRTH, Road Accidents in India 2023, accidents by road feature (reported by PTI, Aug 2025).

How PathPrahari works



Open road data

Curves, junctions, lighting, gradients, blackspots, weather



Fleet events

Harsh braking, near-misses, incidents, GPS, vehicle-km driven



Risk model

Risk for every road segment, by hour and weather, with a 90% range



Options

Fastest, Balanced, Safest routes and the best departure hour



Dispatch and learn

Assign, brief the driver, log the trip, compare after

PathPrahari learns which road stretches are dangerous at which hours from open data and the fleet's own driving, then shows the dispatcher the risk of each option before the truck leaves, with the reasons and how sure it is.

It can also take camera-detected events such as tailgating and distraction, not just harsh braking.

What makes it different



Which way and when

Each stretch is scored for the hour the truck will actually be on it. Departure times are searched along with routes.



Useful from day one

A starting estimate from open data covers every road in India, so a fleet with no history still gets scores. Its own data sharpens them.



Says how sure it is

Every score has a 90% range and a High / Medium / Low label. Roads with little data are flagged, not guessed.



Explains itself in Hindi

Reasons in plain words for the dispatcher and a short Hindi or English briefing for the driver before the trip.

Also: risk is measured per vehicle-km, so a busy highway is not marked dangerous just because it is busy.

The prototype: dispatch console



Screenshot of PathPrahari_Dashboard.html, generated by run_demo.py. It works offline and covers 3 trips × 24 hours × 3 weather conditions.

1. **Set the trip**
   Route, departure hour and weather forecast
2. **See risk on the map**
   Every road coloured by its risk at that hour; hotspots circled
3. **Compare options**
   Time, risk, 90% range and confidence for each route
4. **Read the reasons**
   Top risky stretches with km markers and why
5. **Assign**
   Driver gets the briefing; the trip goes to the log

Which route and what time to leave

Depot A → Hub B, leaving 01:00, clear night

Fastest: straight through on the NH

106 min 6.2

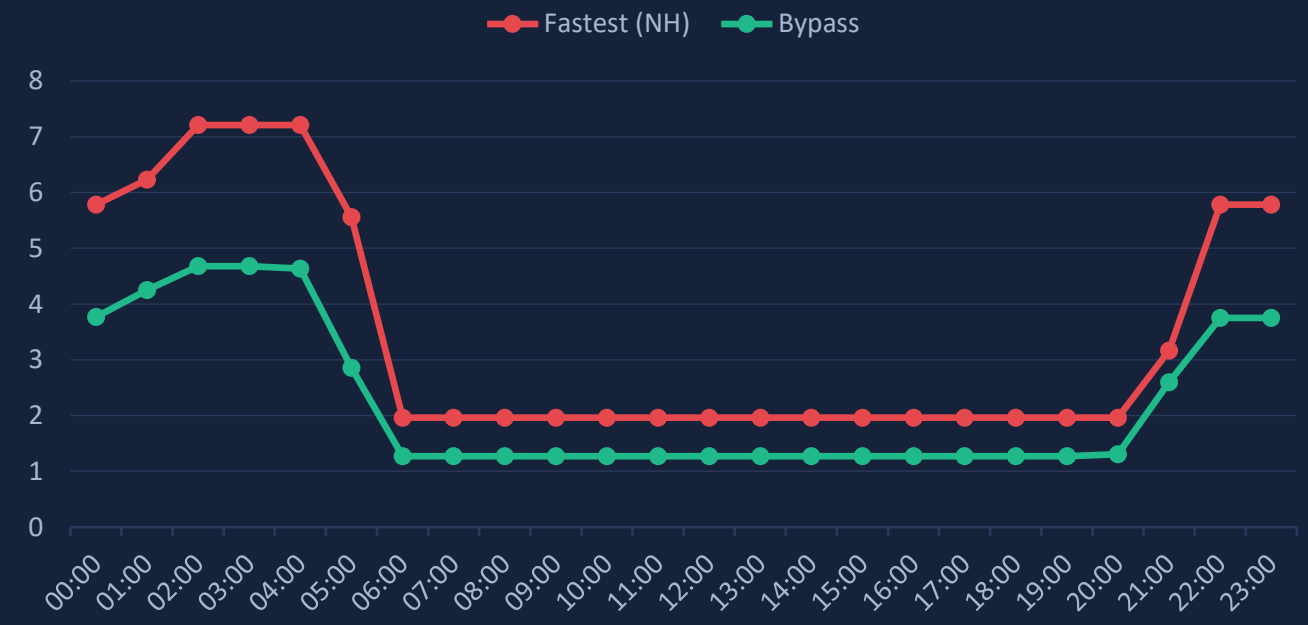
Recommended: state-highway bypass

122 min 4.2

Risk = expected incidents per 10,000 trips

−32% risk for +17 minutes. The bypass avoids a hilly, unlit NH stretch with two fleet hotspots. The 90% ranges (4.0–4.5 vs 6.1–6.4) don't overlap.

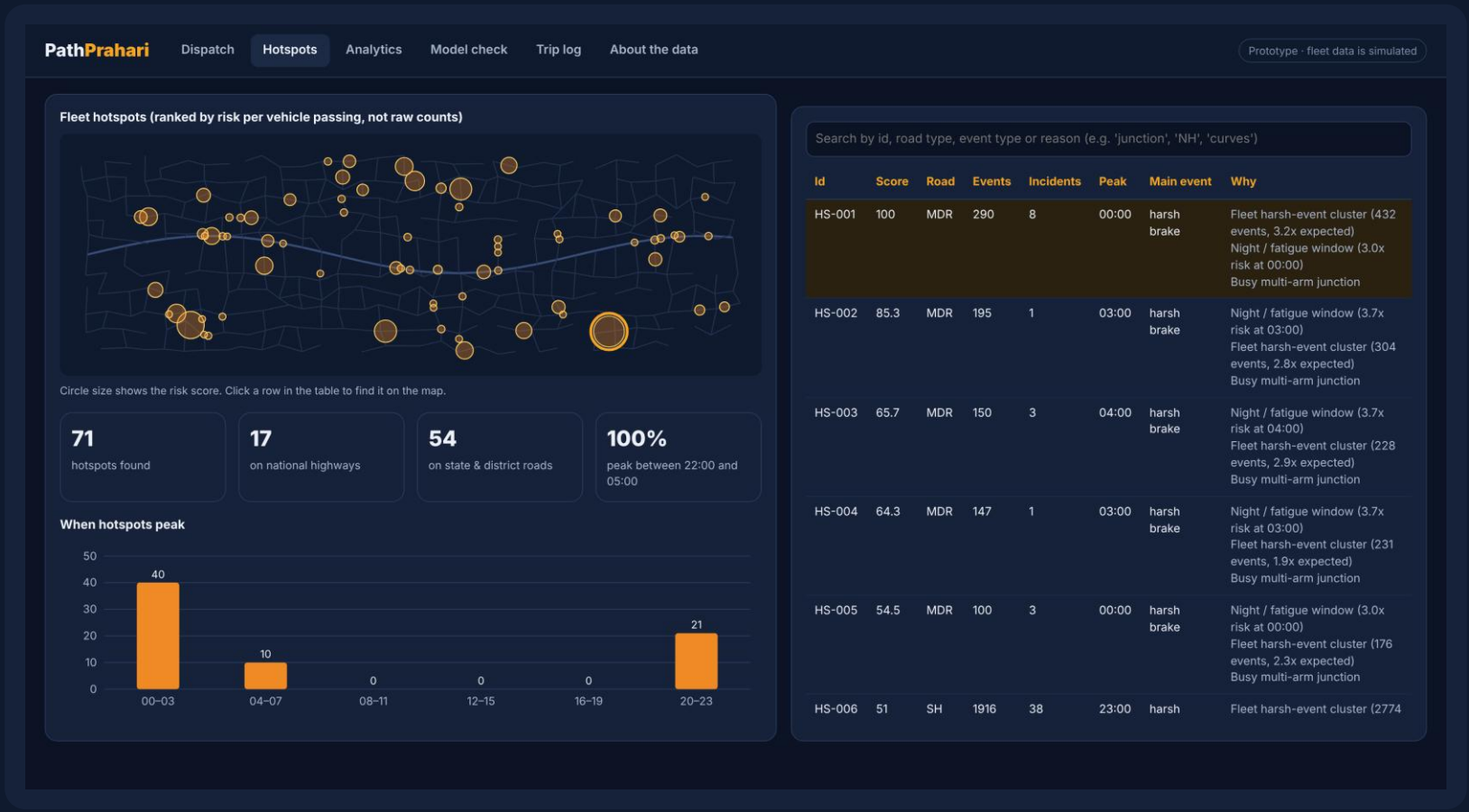
Expected incidents per 10,000 trips, by departure hour



−70% risk on the bypass if the same trip leaves at 06:00 instead of 01:00. It stops crossing unlit, hilly stretches during the night and fatigue hours.

Prototype results on simulated fleet data. Options more than 30% slower than the fastest route are never offered.

Hotspots: where our trucks keep braking hard



Per vehicle, not per count

A busy highway always has more events. Dividing by traffic shows where each truck is most likely to get into trouble.

Each hotspot explains itself

Example from the list: a district-road junction where our trucks brake hard about three times more often than expected, mostly around midnight.

What dispatchers do with it

Route around the worst ones at night, brief drivers about the rest and pass the list to road authorities.

Events are weighted by severity (near-miss counts more than a harsh brake) and by how recent they are, then clustered. The list is searchable in the dashboard.

The risk model

1. Starting estimate

Poisson model on road features: curvature, gradient, junctions, night-light, road type. Covers roads the fleet has never driven.

2. Fleet evidence

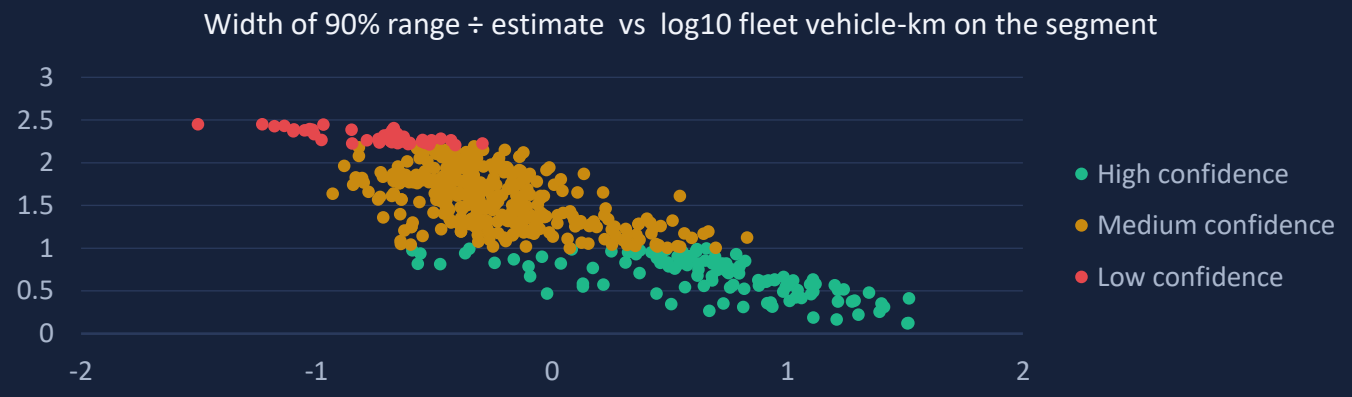
Incidents plus harsh brakes, turns and near-misses on each segment, divided by the vehicle-km driven there.

3. Updated risk

Empirical-Bayes Poisson–Gamma update: more fleet data, narrower range. Gives the mean and a 90% range.

4. Trip score

Multiplied by the learned effect of the hour and weather at each stretch, then summed along the route (Monte Carlo for the range).



Learned from fleet data

Night (22–05h)	×2.63
Fog + night + unlit	×2.64
Fog	×1.48
Rain on curves	×1.39
Fatigue window 02–05h	×1.25

43 of 517 segments are Low confidence

System architecture

1 Open data

OpenStreetMap
roads, curves, junctions

NASA VIIRS
street lighting

SRTM
gradients

MoRTH / data.gov.in
blackspots, crash stats

Open-Meteo / IMD
rain, fog, visibility

2 Fleet data

Telematics feed
events + GPS + time

Map-matching
segments + H3 cells

Exposure ledger
vehicle-km by hour, weather

Cleaning
GPS jumps, duplicates

3 Intelligence

Hotspot engine
weighted, traffic-normalised

Segment risk
prior + empirical Bayes

Context model
hour, fatigue, fog, rain

Route + time optimiser
trade-off curve, best hour

Reason codes
plain-language reasons

4 Delivery

Dispatch console
map, options, assign

REST API
for transport software

Driver briefing
Hindi / English

Reports
trends, hotspots, trip log

↻ **Learning loop:** every dispatch is logged with its score → compared with harsh events and incidents on the trip → model refits monthly.
Fleets can opt in to share anonymised hotspot counts.

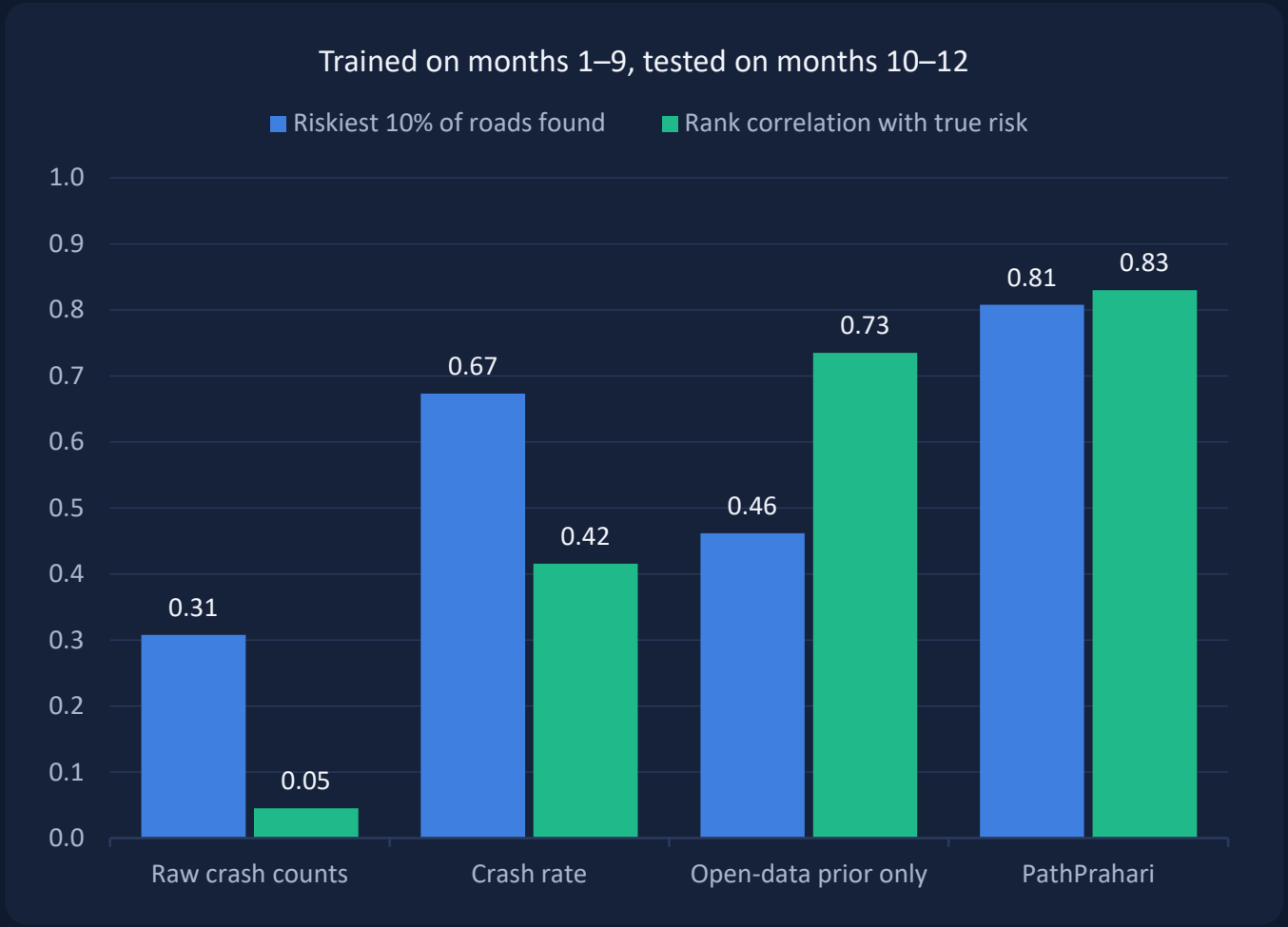
Built with Python, NetworkX, scikit-learn, SciPy, pandas. Real-data loaders: OSMnx, GeoPandas, H3. Service: FastAPI + PostGIS. MIT licence.

Where the data comes from

Source	What PathPrahari uses it for	Licence	In the prototype
OpenStreetMap (Geofabrik India)	Road network, curvature, junctions, road type, speed	ODbL 1.0	Same fields, generated network
NASA Black Marble VNP46A2	Night-light radiance as a street-lighting index	NASA open data	Same field, generated values
SRTM 30 m elevation (USGS)	Gradients and hilly sections	Public domain	Same field, generated hills
MoRTH Road Accidents in India 2023	Problem sizing; calibrating starting rates by state and road type	Govt. open data	Problem figures
MoRTH NH blackspot lists	Blackspot flag; held-out check of the model	Govt. publication	Planned validation
Open-Meteo / IMD	Rain, fog and visibility, past and forecast	CC BY 4.0 / IMD terms	Fog and rain seasons simulated
Fleet telematics	Harsh events, near-misses, incidents with GPS	Owned by the fleet	Simulated: 180 trucks, 12 months

Links, refresh cadence and licence terms are in DATA_SOURCES.csv. No personal data is used. Fleet data stays with the fleet unless it chooses to share aggregated hotspot counts.

Testing it on months it never saw



66% lower prediction error than ranking roads by crash counts

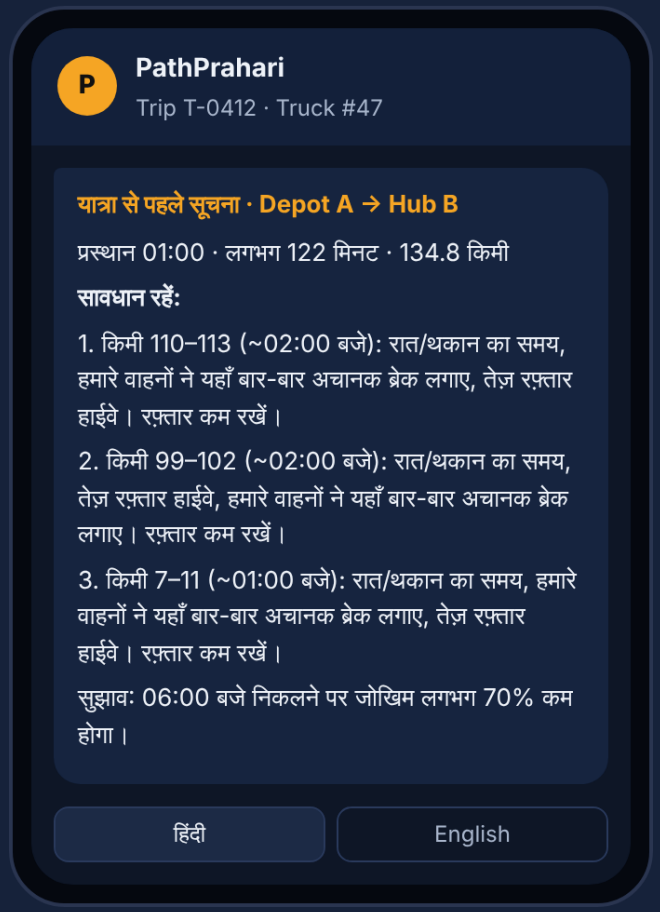
97% of roads had their true risk inside the 90% range

0.99 rank agreement with true route risk over 239 random trips

5.1M events processed in 6 s on one laptop core

Simulated data, so the true risk of every road is known and accuracy can be measured exactly. A pilot would repeat these checks on a real fleet.

Getting it to the driver and learning from every trip



A briefing the driver can actually use

- The three riskiest stretches, with km markers and the time the truck will reach them
- In Hindi or English, as text now and voice next; more languages after the pilot
- Written from the same reasons the dispatcher sees, so both are on the same page

Closing the loop (120 simulated dispatches)

63 → 32 harsh events per trip: fastest route vs the recommended one, counted after the trip

Pre-trip score vs harsh events after the trip: rank correlation 0.91. Each month's trips feed the next refit.

Who it helps and how we'll measure it

Drivers

Fewer night runs through dark, hilly stretches; warned about the exact km where trouble is likely.

Fleet owners

Fewer crashes and claims, less downtime and a record to show insurers and customers.

Other road users

Fewer heavy vehicles on risky stretches at the worst hours, on the roads with the most deaths.

Road authorities

Fleet-sourced hotspot lists that show trouble spots before they reach blackspot status.

What we expect

Fleet risk sits mostly on a small share of stretches and night hours. Avoiding those only when it costs little time should cut incidents and harsh events per 1,000 trips while delivery times barely change.

This is a hypothesis we will test, not a promise: the pilot compares the same fleet before and after suggestions are switched on.

What we'll measure in the pilot

Incidents per million vehicle-km · main outcome

Harsh events per trip · early signal

Share of trips on safer options · whether dispatchers use it

Minutes added per trip · what safety costs

How we take it forward

1

Weeks 0–4

Open-source release

Loaders for real OSM, VIIRS and SRTM data; demo on the Varanasi–Prayagraj–Lucknow corridor

Done when: Any corridor scored with one command

2

Months 2–5

Pilot with one fleet

Use the fleet's existing GPS export. Score trips quietly for 3 months, then switch on suggestions

Done when: KPIs measured against the quiet period

3

Months 5–8

Connect and scale

API plugins for transport-management software; driver briefings through chat apps; more languages

Done when: Two integrations live

4

Months 8–12

Shared hotspot map

Fleets opt in to share anonymised hotspot counts; reports offered to road authorities

Done when: 3+ fleets sharing; first authority report

What we need: one pilot fleet willing to share 6–12 months of GPS and event data and a dispatcher team to try the console. A single cloud VM is enough to run it.

What could go wrong and our plan for it

Telematics data is patchy

GPS gaps and noisy events on some trucks.

Cleaning and map-matching; thin data widens the 90% range and lowers confidence instead of producing a confident wrong number.

Dispatchers ignore it

A tool that slows them down will not be used.

Quiet scoring first, then suggestions with the time cost shown upfront. The dispatcher always decides. Adoption is a pilot KPI.

Roads change

A blackspot gets fixed, a new bypass opens.

Recent events count more (3-month half-life) and the model refits monthly.

Only driven roads have data

Past routes bias what the fleet knows.

The open-data estimate covers every road; undriven roads are marked Low confidence.

Privacy

Driver and fleet data are sensitive.

Data stays with the fleet. Only anonymised, aggregated hotspot counts are ever shared and only if the fleet opts in.

Crashes are rare

Too few incidents to learn from alone.

Harsh events and near-misses are used as early signals, scaled to match incident rates.

PathPrahari

A safer route or a safer hour, decided before the truck leaves.

- Scores routes for the hour the truck is actually on each stretch and finds better departure times
- Works on any Indian road from open data and improves with each fleet's own trips
- Shows how sure it is, so dispatchers know when to trust a score
- Gives reasons a dispatcher can check and a briefing a driver can follow
- Open-source, with a working prototype and a clear pilot plan

Thank You